ST. GREGORIOS SCHOOL, DWARKA

MID-TERM EXAMINATION (2026-27)

MATHEMATICS (041)

Date: 09.09.26

Max. Marks: 80

CLASS – X

Time: 3 hours

General Instructions:

Read the following instructions very carefully and strictly follow them:

1. This question paper contains 38 questions. All Questions are compulsory.

2. This question paper is divided into 5 sections - Section A, B, C, D and E.

3. In Section A, question numbers 1 to 18 are multiple choice questions (MCQs) and questions numbers 19 and 20 are Assertion- Reason based questions of 1 mark each.

4. In Section B, question numbers 21 to 25 are very short answer (VSA) type questions, carrying 2 marks each.

5. In Section C, question numbers 26 to 31 are short answer (SA) type questions, carrying 3 marks each.

6. In Section D, question numbers 32 to 35 are long answer (LA) type questions, carrying 5 marks each.

7. In Section E, question numbers 36-38 are case study-based questions, carrying 4 marks each. Internal choice is provided in 2 marks question in each case study.

8. There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 2 questions in Section C, 2 questions in Section D and 3 questions of 2 marks in Section E.

9. Draw neat figures wherever required. Take $\pi = 22/7$ wherever required, if not stated.

SECTION - A (20x1=20)

1. The product of a non-zero rational and an irrational number is

a) always irrational b) always rational c) rational or irrational d) one

2. If two positive integers a and b are written as $a=x^3y^2$ and $b=xy^3$; x ,y are prime numbers, then HCF(a, b) is

(a) xy (b) xy^2 (c) x^3y^3 (d) xy^3

3. If one zero of the quadratic polynomial $(k-1)x^2 + kx + 1$ is -3, then the value of k is

(a) 4/3 (b) -4/3 (c) 2/3 (d) -2/3

4. A pair of linear equations $a_1x + b_1y + c_1 = 0$; $a_2x + b_2y + c_2 = 0$ is said to be *inconsistent, if*

(a) $\frac{a1}{a2} \neq \frac{b1}{b2}$ (b) $\frac{a1}{a2} = \frac{b1}{b2} \neq \frac{c1}{c2}$ (c) $\frac{a1}{a2} = \frac{b1}{b2} = \frac{c1}{c2}$ (d) None of these

5. The pair of equations y=0 and y=(-7) graphically represents lines which have

(a) one solution (b) two solutions (c) no solution (d) infinitely many solutions

6. One equation of a pair of dependent linear equations is $-5x + 7y = 2$. The second equation can be

(a) 10x+14y+4=0 (b) -10x-14y+4=0

(c) -10x+14y+4=0 (d) 10x – 14y= -4

7. The graph of the polynomial $ax^2 + bx + c$ is an upward parabola if

(a) $a > 0$ b) $a < 0$ c) $a = 0$ d) $a = 1$

8. The quadratic equation $x^2 - 3\sqrt{5}x + 10 = 0$ has

(a) two distinct real roots (b) two equal real roots

(c) no real roots (d) more than 2 real roots

9. Which of the following equations has the sum of the roots as 3?

(a) $2x^2 - 3x + 6 = 0$ (b) $-x^2 + 3x - 3 = 0$ (c) $2x^2 - 3x + \sqrt{2} = 0$ (d) $3x^2 - 3x + 3 = 0$

10. The distance (in units) of the point P(2,3) from the x-axis is

(a) 2 (b) 3 (c)5 (d)1

11. The points A(-4,0), B(4, 0) and C(0,3) are the vertices of a

(a) right triangle (b) isosceles triangle (c) equilateral triangle (d) scalene triangle

12. The total surface area of a hemispherical solid having diameter 7 cm is

(a) 77 cm^2 (b) 115.5 cm^2 (c) 462 cm^2 (d) 154 cm^2

13. Volume of two spheres are in the ratio 64: 27. The ratio of their surface areas is

(a) 3: 4 (b) 4 : 3 (c) 9 : 16 (d) 16 : 9

14. If the area of the base of a cone is 5.1 m^2 and its volume is 10.2 m^3, then

the vertical height of the cone is given as:

(a) 5 m (b) 6.5 m (c) 4 m (d) 6 m

15. The empirical relation between the mode, median and mean of a distribution is :

(a) Mode = 2 Median – 3Mean (b) Mode = 3 Mean – 2 Median

(c) Mode = 3 Median – 2 Mean (d) Mode = 2 Mean – 3 Median

16. What is the area of a semi-circle of diameter 'd'?

(a) $\frac{1}{16}\pi d^2$ (b) $\frac{1}{4}\pi d^2$ (c) $\frac{1}{8}\pi d^2$ (d) $\frac{1}{2}\pi d^2$

17. For the following distribution:

Class	10-20	20-30	30-40	40-50	50-60
Frequency	15	20	9	5	11

The sum of upper limits of median class and modal class is:

(a) 40 (b) 50 (c) 60 (d) 70

18. In a lucky draw, there are 5 prizes and 20 blanks. The probability of getting a prize is:

(a) $\frac{1}{4}$ (b) $\frac{1}{20}$ (c) $\frac{1}{25}$ (d) $\frac{1}{5}$

Questions number 19 and 20 are Assertion and Reason based questions. Two statements are given, one labelled as Assertion (A) and the other is labelled as Reason (R). Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

19. Assertion (A): The probability of losing a game is 0.65, then the probability of winning it is 0.45

Reason (R): For any event E, P(E)+P (not E) = 1.

20. Assertion (A): If the radius of a sphere is doubled, its volume increases by a factor of 8.

Reason (R): The total surface area of a sphere is directly proportional to the square of its radius.

SECTION -B (5x2=10)

21. Prove that 6^n can never end with digit 0, where n is a natural number.

OR

Find the HCF and LCM of 96 and 404 by the prime factorization method.

22. A coin is tossed two times. Find the probability of getting atmost one head.

23. For which values of a and b, will the following pair of linear equations have infinitely many solutions?

$x+2y=1$, $(a-b)\,x + (a+b)\,y = a + b - 2$

24. Find the roots of $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$

OR

Find two consecutive positive integers, sum of whose squares is 365.

25. A bag contains 10 red, 5 blue and 7 green balls. A ball is drawn at random, find the probability of this ball being a (i) red ball (ii) not a blue ball

SECTION - C (6x3=18)

26. A train travels a distance of 480 km at a uniform speed. If the speed had been 8 km/h less, then it would have taken 3 hours more to cover the same distance. Find the speed of the train.

27. Find the zeroes of the quadratic polynomial, $4x^2 - 3x - 1$ and verify the relationship between the zeroes and the coefficients.

28. Solve the following pair of equations.

$\frac{3x}{2} - \frac{5y}{3} = -2$, $\frac{x}{3} + \frac{y}{2} = \frac{13}{6}$

OR

In a chemistry lab, there is some quantity of 50% acid solution and some quantity of 25% acid solution. How much of each should be mixed to make 10 litres of 40% acid solution?

29. Find the ratio in which the y-axis divides the line segment joining the points (5,-6) and (-1, -4). Also find the point of intersection.

OR

Find the ratio in which the point $(\frac{8}{5}, y)$ divides the line segment joining the points (1,2) and (2,3). Also, find the value of y.

30. A vessel is in the form of a hollow hemisphere mounted by a hollow cylinder. The radius of the hemisphere is 7 cm the total height of the vessel is 13 cm. Find the inner surface area of the vessel.

31. Find the mode of the following data where total frequency is 50.

Class Interval	100-110	110-120	120-130	130-140	140-150
Frequency	4	14	21	8	3

SECTION – D **(4x5=20)**

32. Prove that $\sqrt{3}$ is irrational and also prove that $2 - 5\sqrt{3}$ is also irrational.

33. Graphically, solve the following pair of equations:

$2x+y=6$, $2x - y+2=0$

Determine the coordinates of the vertices of the triangle formed by these lines and the y-axis and also find the area of the triangle so formed.

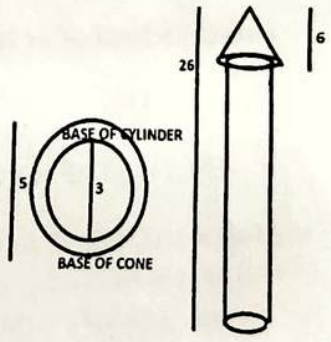
34. Water in a canal, 6m wide and 1.5 m deep, is flowing with a speed of 10 km/h.

(i) Calculate the volume of water that flows through the canal in 30 minutes.

(ii) How much area will this water irrigate in 30 minutes, if 8 cm of standing water is required for irrigation?

OR

A wooden toy rocket is in the shape of a cone mounted on a cylinder, as shown in Fig. The height of the entire rocket is 26 cm, while the height of the conical part is 6 cm. The base of the conical portion has a diameter of 5 cm, while the base diameter of the cylindrical portion is 3 cm. If the conical portion is to be painted orange and the cylindrical portion yellow, find the area of the rocket painted with each of these colours. (**Take $\pi = 3.14$**)



35. If the mean of the distribution given below is 57.6, find the values of x and y.

C.I.	0-20	20-40	40-60	60-80	80-100	100-120	*Total*
Frequency	7	x	12	y	8	5	**50**

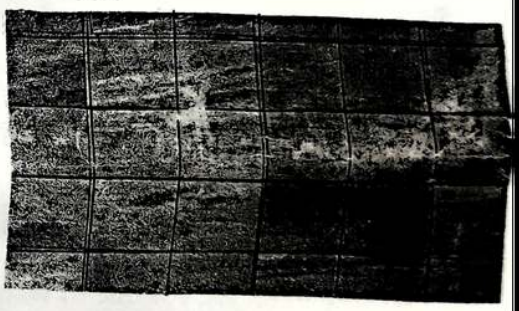
OR

If the median of the distribution given below is 35, find the values of x and y.

C.I.	10-20	20-30	30-40	40-50	50-60	60-70	**Total**
Frequency	6	x	12	y	3	1	**40**

SECTION-E **(3X4=12)**

36. A rectangular floor area can be completely tiled with 200 square tiles. If the side length of each tile is increased by 1 unit, it would take only 128 tiles to cover the floor.



(i) Assuming the original length of each side of a tile be x units, make a quadratic equation from the above information.

(ii) Write the corresponding quadratic equation in standard form.

(iii) (a) Find the value of x, the length of side of a tile by factorisation.

OR

(b) Solve the quadratic equation for x, using quadratic formula.

37. Two dice are thrown simultaneously. Find the probability of getting:

(i) an odd number as the sum

(ii) a doublet of odd number.

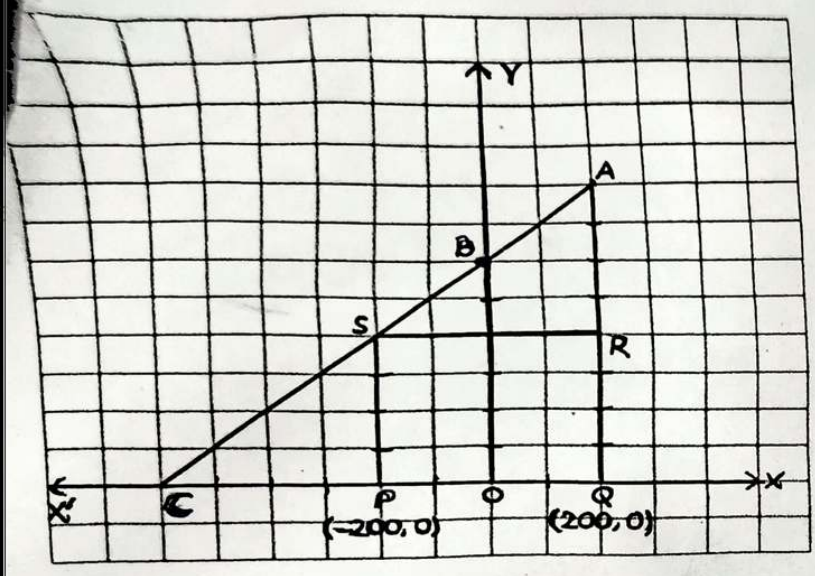


(iii) (a) a total of at least 9

OR

(b) a total of at most 7

38. Jagan has a field in the shape of a right-angled triangle AQC. He wants to leave a square PQRS inside the field for growing wheat and the remaining for growing vegetables, as shown in the figure. There is a pole marked as O.



(i) Taking O as origin, coordinates of P are (-200, 0) and of Q are (200, 0).

PORS being a square, what are the coordinates of R and S?

(ii) If S divides CA in the ratio k :1, what is the value of k, where point A is (200, 800)?

(iii) (a) What is the area of square PQRS?

OR

(b) What is the length of diagonal PR in square PQRS?

— x —